

Contour Integration

Application of Residues Theorem

2π [Contour Integration - First Type]

• $\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$, C is a circle with unit radius,

where F is a rational function.

$$z = e^{i\theta}$$

$$dz = i e^{i\theta} d\theta = iz d\theta$$

$$d\theta = \frac{dz}{iz}$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{z + z^{-1}}{2}$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} = \frac{z - z^{-1}}{2i}$$

$$\int_C F\left(\frac{z + z^{-1}}{2}, \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}$$

$$= 2\pi i (\text{Sum of residues of } f(z) \text{ at the poles lying inside } C.)$$

Q. Evaluate the following integration by contour integration:

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta$$

Solⁿ:

We have, $e^{i3\theta} = \cos 3\theta + i \sin 3\theta$

Let, $z = e^{i\theta} \Rightarrow z^3 = e^{i3\theta}$

$$dz = i e^{i\theta} d\theta \Rightarrow d\theta = \frac{dz}{iz}$$
$$= iz d\theta$$

$$\cos \theta = \frac{z + z^{-1}}{2} = \frac{z + 1/z}{2} = \frac{z^2 + 1}{2z}$$

R.P. of

$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta = \text{R.P. of } \int_0^{2\pi} \frac{e^{i3\theta}}{5 - 4 \cos \theta} d\theta$$

$$= \text{R.P. of } \int_C \frac{z^3}{5 - 4 \left(\frac{z^2 + 1}{2z} \right)} \frac{dz}{iz}$$

$$= \text{R.P. of } \int_C \frac{z^3 (2z)}{10z - 4z^2 - 4} \cdot \frac{dz}{i}$$

$$= \text{R.P. of } \int_C \frac{2z^3}{i(10z - 4z^2 - 4)} dz$$

$$= \text{R.P. of } \frac{1}{i} \int_C \frac{2z^3}{-(4z^2 - 10z + 4)} dz$$

$$= \text{R.P. of } \cancel{i} \int_C \cancel{-} \frac{2z^3}{(4z^2 - 10z + 4)} dz$$

$$\left[\frac{1}{i} = \frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i \right]$$

$$= \text{R.P. of } i \int_C \frac{2z^3}{4z^2 - 10z + 4} dz$$

$$\text{Let, } f(z) = \frac{2z^3}{4z^2 - 10z + 4} = \frac{2z^3}{2(2z^2 - 5z + 2)}$$

$$= \frac{z^3}{(z-2)(2z-1)}$$

For poles of $f(z)$,

$$(z-2)(2z-1) = 0$$

$$\Rightarrow z = 2, \frac{1}{2}$$

$z = \frac{1}{2}$ lies inside circle C , $|z| = 1$

$z = \frac{1}{2}$ is simple pole.

$$f(z) = \frac{g(z)}{h(z)}$$

Here, $g(z) = z^3 \Rightarrow g(1/2) = 1/8 \neq 0$
&

$$h(z) = (z-2)(2z-1) \Rightarrow h(1/2) = 0$$

So, $h'(z) = 2(z-2) + (2z-1)$

$$\Rightarrow h'(1/2) = 2\left(\frac{1}{2} - 2\right) + \left(2 \cdot \frac{1}{2} - 1\right)$$

$$= 2\left(-\frac{3}{2}\right) + 0$$

$$= -3$$

$$\therefore \text{Res}(z=1/2) = \frac{g(1/2)}{h'(1/2)} = \frac{1/8}{-3} = -\frac{1}{24}$$

Now,

$$\int_C \frac{z^3}{2z^2 - 5z + 2} dz = 2\pi i \left(\frac{-1}{24} \right) = -\frac{\pi i}{12}$$

Thus, $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta} d\theta = \text{R.P. of } \int_0^{2\pi} \frac{e^{3i\theta}}{5-4\cos\theta} d\theta$

$$= \text{R.P. of } i \int \frac{z^3}{2z^2 - 5z + 2} dz$$

$$= \text{R.P. of } i \left(\frac{-\pi i}{12} \right) = \frac{\pi}{12}$$

$$= \text{R.P. of } \left(\frac{\pi}{12} \right)$$

$$\therefore \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos \theta} d\theta = \frac{\pi}{12}$$

Q. Evaluate $\int_0^{2\pi} \frac{d\theta}{a + b \sin \theta}$, $a > |b|$.

Solⁿ:

Let, $z = e^{i\theta} \Rightarrow dz = i e^{i\theta} d\theta \Rightarrow dz = iz d\theta$

$$d\theta = \frac{dz}{iz}$$

$$\sin \theta = \frac{z - z^{-1}}{2i} = \frac{z - 1/z}{2i} = \frac{z^2 - 1}{2iz}$$

$$\int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \int_C \frac{1}{a + b \left(\frac{z^2 - 1}{2iz} \right)} \cdot \frac{dz}{iz}$$

$$= \int_C \frac{\cancel{2iz}}{2iaz + bz^2 - b} \cdot \frac{dz}{\cancel{iz}}$$

$$= 2 \int_C \frac{dz}{bz^2 + 2iaz - b}$$

$$= \frac{2}{b} \int_C \frac{dz}{z^2 + \frac{2iaz}{b} - 1}$$

$$f(z) = \frac{1}{z^2 + \frac{2iaz}{b} - 1}$$

For poles of $f(z)$,

$$z^2 + \frac{2iaz}{b} - 1 = 0$$

$$z = \frac{-\frac{2ai}{b} \pm \sqrt{\frac{4i^2a^2}{b^2} + 4}}{2}$$

$$\Rightarrow z = \frac{-\frac{2ai}{b} \pm \frac{2}{b} \sqrt{-a^2 + b^2}}{2}$$

$$\Rightarrow z = \frac{-ai \pm \sqrt{-a^2 + b^2}}{b}$$

$$\Rightarrow z = \frac{-ai}{b} \pm \frac{\sqrt{-a^2 + b^2}}{b}$$

$$\text{Let, } \alpha = \frac{-ai}{b} + \frac{\sqrt{-a^2 + b^2}}{b} = -\left(\frac{ai}{b} - \frac{\sqrt{-a^2 + b^2}}{b}\right)$$

$$\& \beta = \frac{-ai}{b} - \frac{\sqrt{-a^2 + b^2}}{b} = -\left(\frac{ai}{b} + \frac{\sqrt{-a^2 + b^2}}{b}\right)$$

$$\text{Given, } a > |b| \Rightarrow \left|\frac{a}{b}\right| > 1$$

$$\text{So, } |\alpha| < 1 \& |\beta| > 1$$

$$\text{So, } z = \alpha \text{ lies inside } C \quad |z| = 1.$$

$\therefore z = \alpha$ is a simple pole.

$$\therefore \text{Res}(z = \alpha) = \lim_{z \rightarrow \alpha} \left[\cancel{(z - \alpha)} \frac{1}{\cancel{(z - \alpha)}(z - \beta)} \right]$$

$$= \lim_{z \rightarrow \alpha} \left(\frac{1}{z - \beta} \right)$$

$$= \frac{1}{\alpha - \beta}$$

Now,

$$\alpha - \beta = \cancel{\frac{-ai}{b}} + \frac{\sqrt{-a^2 + b^2}}{b} + \cancel{\frac{ai}{b}} + \frac{\sqrt{-a^2 + b^2}}{b}$$

$$\Rightarrow \alpha - \beta = \frac{2\sqrt{-a^2 + b^2}}{b}$$

$$\text{So, Res}(z = \alpha) = \frac{b}{2\sqrt{-a^2 + b^2}}$$

$$\text{Then, } \int_C \frac{dz}{(z - \alpha)(z - \beta)} = 2\pi i \left(\frac{b}{2\sqrt{-a^2 + b^2}} \right)$$

$$\therefore \int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \frac{2}{b} \left[\frac{\pi i b}{\sqrt{-a^2 + b^2}} \right] = \frac{2\pi i}{\sqrt{-a^2 + b^2}}$$

$$= \frac{2\pi}{\sqrt{a^2 - b^2}}$$

$$\left[\sqrt{-a^2 + b^2} = \sqrt{-(a^2 - b^2)} = \sqrt{(-1)} \cdot \sqrt{a^2 - b^2} = \sqrt{i^2} \sqrt{a^2 - b^2} = i\sqrt{a^2 - b^2} \right]$$

Contour Integration - Second Type

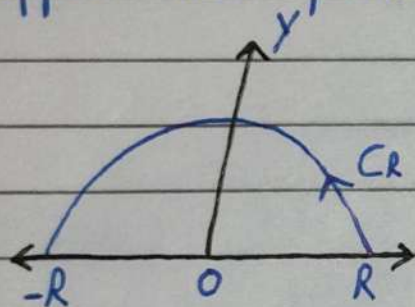
Integral of type $\int_{-\infty}^{\infty} \frac{f(x)}{F(x)} dx$, where $f(x)$ and $F(x)$

are polynomials in x such that $x \frac{f(x)}{F(x)} \rightarrow 0$

as $x \rightarrow \infty$ and $F(x)$ has no zeroes on the real axis.

Consider the integral $\oint_C \frac{f(z)}{F(z)} dz$ over the closed

curve C consisting of real axis from $-R$ to R and the semi-circle C_R of radius R in the upper half plane.



$$\oint_C \frac{f(z)}{F(z)} dz = 2\pi i \left(\text{Sum of residues of } \frac{f(z)}{F(z)} \text{ in the upper half plane} \right)$$

$$\Rightarrow \oint_{C_R} \frac{f(z)}{F(z)} dz + \int_{-R}^R \frac{f(x)}{F(x)} dx = \oint_C \frac{f(z)}{F(z)} dz \quad (1)$$

[On the real axis $z=x$]

$$\text{Let, } z = R e^{i\theta} \Rightarrow dz = R e^{i\theta} i d\theta$$
$$\Rightarrow dz = R i z d\theta$$

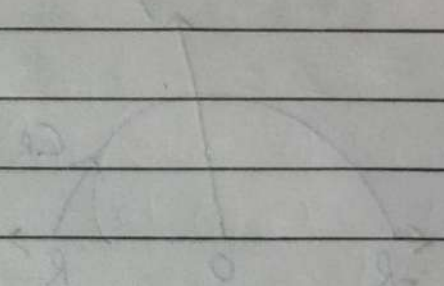
$$\int_{C_R} \frac{f(z)}{F(z)} dz = \int_0^\pi \frac{f(R e^{i\theta})}{F(R e^{i\theta})} \cdot R e^{i\theta} i d\theta$$

For large R ,

$$\left| \int_0^\pi \frac{f(Re^{i\theta})}{F(Re^{i\theta})} Re^{i\theta} i d\theta \right| \rightarrow 0$$

So, $\int_0^\pi \frac{f(Re^{i\theta})}{F(Re^{i\theta})} Re^{i\theta} i d\theta \rightarrow 0$ when $R \rightarrow \infty$

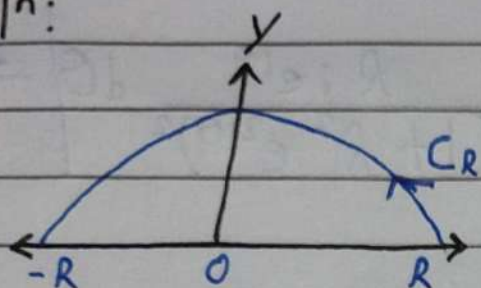
$$\therefore \int_{-\infty}^{\infty} \frac{f(x)}{F(x)} dx = 2\pi i \left(\text{Sum of residues of } f(x)/F(x) \text{ in upper half of the plane.} \right)$$



Q. Using Contour Integration, prove that:

$$\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{2}$$

Solⁿ:



$$\frac{f(x)}{F(x)} = \frac{1}{(1+x^2)^2}$$

Zeros of $F(x)$ are $(1+x^2)^2 = 0$

$\Rightarrow x = \pm i$, which does not lie on real axis.

~~Now, $\frac{x f(x)}{F(x)} = \frac{x}{(1+x^2)^2} = \frac{x}{x^4 (1+1/x^2)^2}$~~

Let, $\int_C \frac{f(z)}{F(z)} dz = \int_C \frac{dz}{(1+z^2)^2}$

$$= \int_{C_R} \frac{dz}{(1+z^2)^2} + \int_{-R}^R \frac{dx}{(1+x^2)^2}$$

$$= 2\pi i \left(\text{Sum of residues of } \frac{1}{(1+z^2)^2} \text{ lies in the upper half of the plane.} \right)$$

Let, $z = R e^{i\theta} \Rightarrow dz = R i e^{i\theta} d\theta$

$$\int_{C_R} \frac{dz}{(1+z^2)^2} = \int_0^\pi \frac{R i e^{i\theta} d\theta}{(1+R^2 e^{2i\theta})^2} \neq \int_0^\pi \frac{R i e^{i\theta} d\theta}{R^4 \left(e^{2i\theta} + \frac{1}{R^2} \right)^2}$$

When $R \rightarrow \infty$ then $\left| \int_0^\pi \frac{R i e^{i\theta}}{(1+R^2 e^{2i\theta})^2} d\theta \right| \Rightarrow 0$

$$\Rightarrow \int_{C_R} \frac{dz}{(1+z^2)^2} \rightarrow 0 \text{ as } R \rightarrow \infty$$

$$\Rightarrow \int_C \frac{dz}{(1+z^2)^2} = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2} = 2\pi i \left(\text{Sum of residues of } \frac{1}{(1+z^2)^2} \text{ at } \right)$$

the poles which lies in upper half plane.

For poles,

$$(1+z^2)^2 = 0$$

$$\Rightarrow z = \pm i, \pm i$$

$$\Rightarrow z = i, i, -i, -i$$

$\therefore z = i$ lies inside C_R and is pole of order 2.

$$\therefore \text{Res}(z=i) = \frac{1}{(2-1)!} \left[\frac{d^{2-1}}{dz^{2-1}} \left\{ \frac{1}{(z-i)^2 (z+i)^2} \right\} \right]_{z=i}$$

$$= \frac{1}{1!} \left[\frac{d}{dz} \frac{1}{(z+i)^2} \right]_{z=i}$$

$$= \left[\frac{-2}{(z+i)^3} \right]_{z=i}$$

$$= \frac{-2}{(i+i)^3}$$

$$= \frac{-2}{(2i)^3}$$

$$= \frac{-2}{8(i)}$$

$$= \frac{1}{4i}$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2} = 2\pi i \left(\frac{1}{4i} \right) = \frac{\pi}{2} \quad \text{proved.}$$